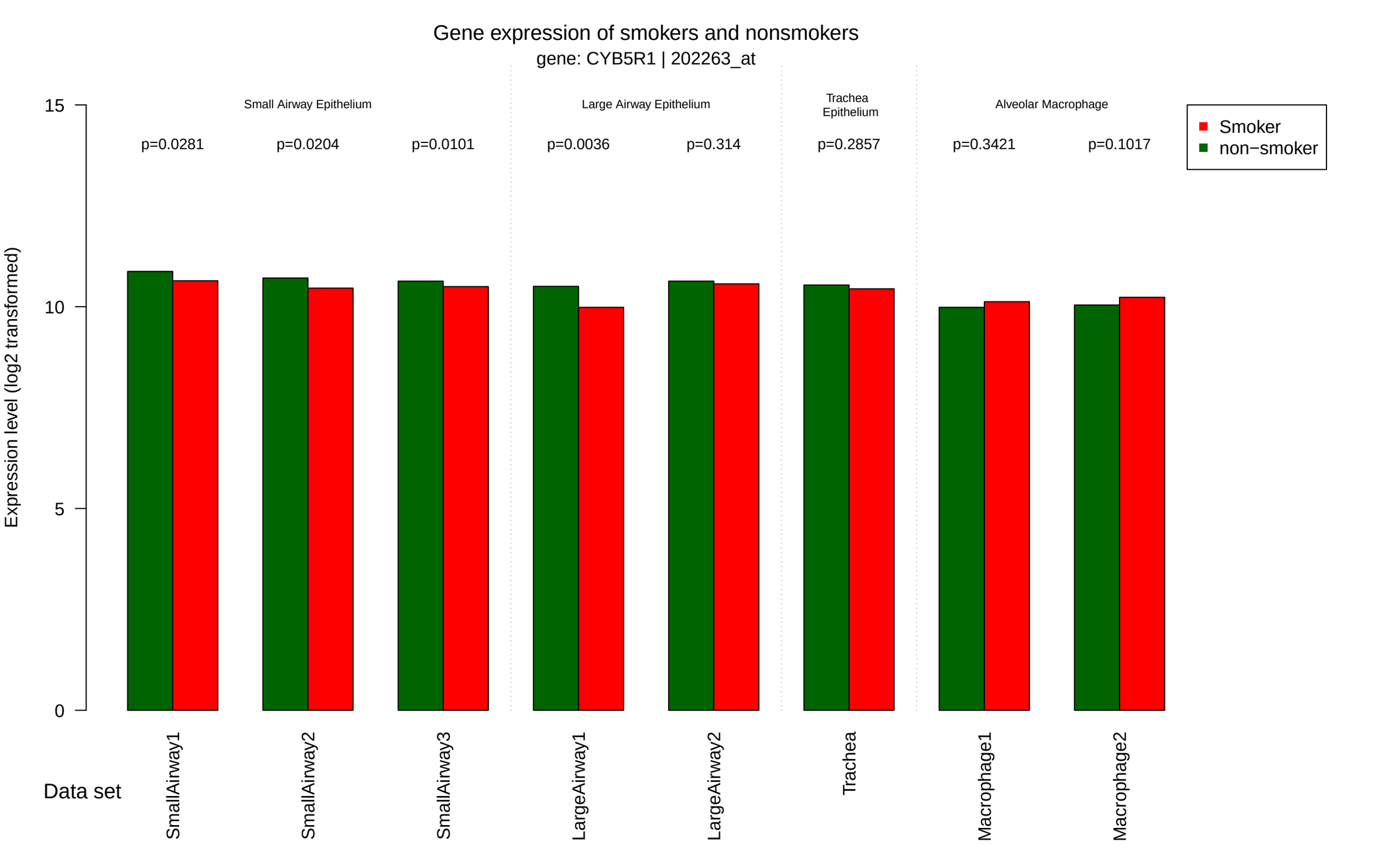
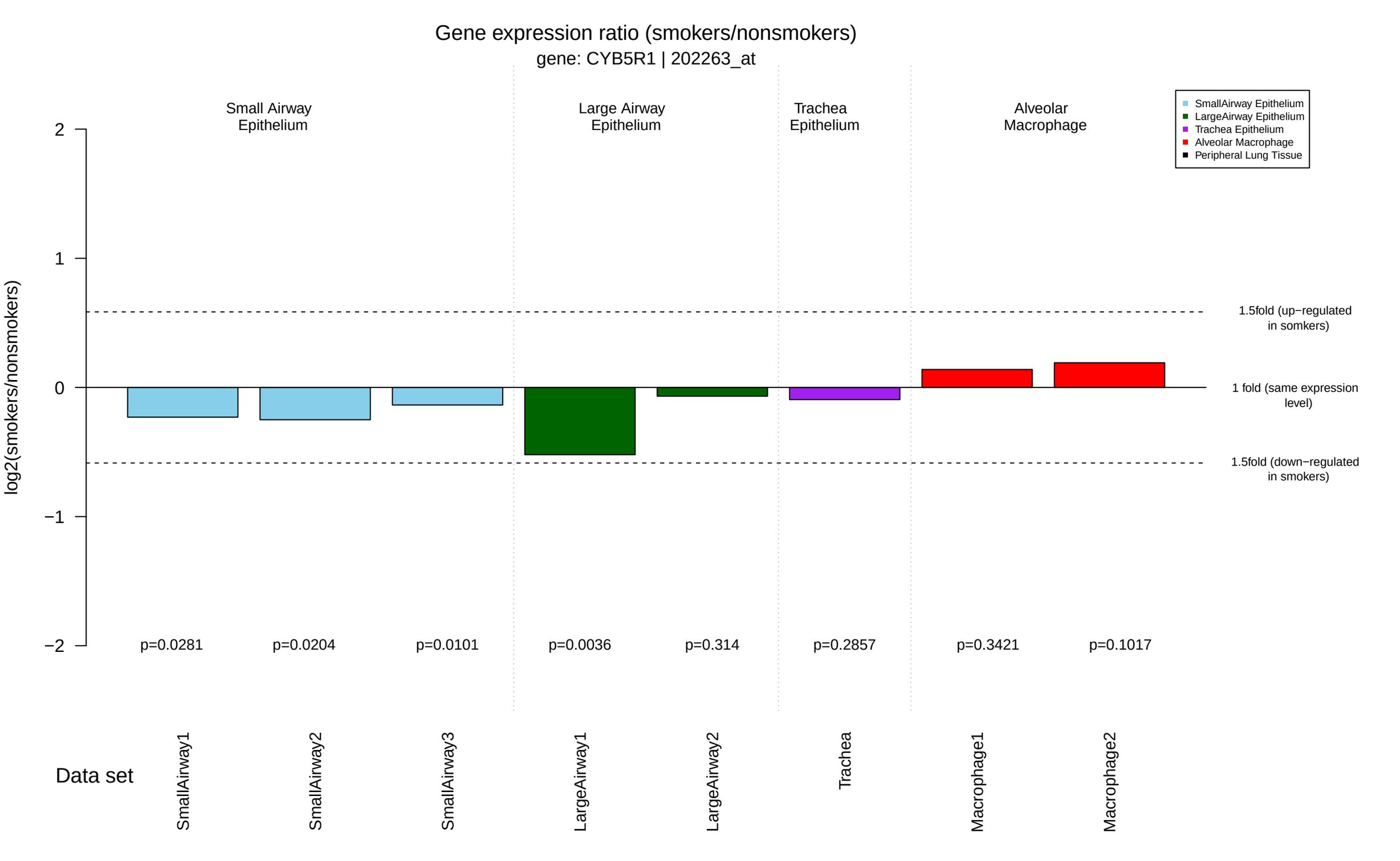
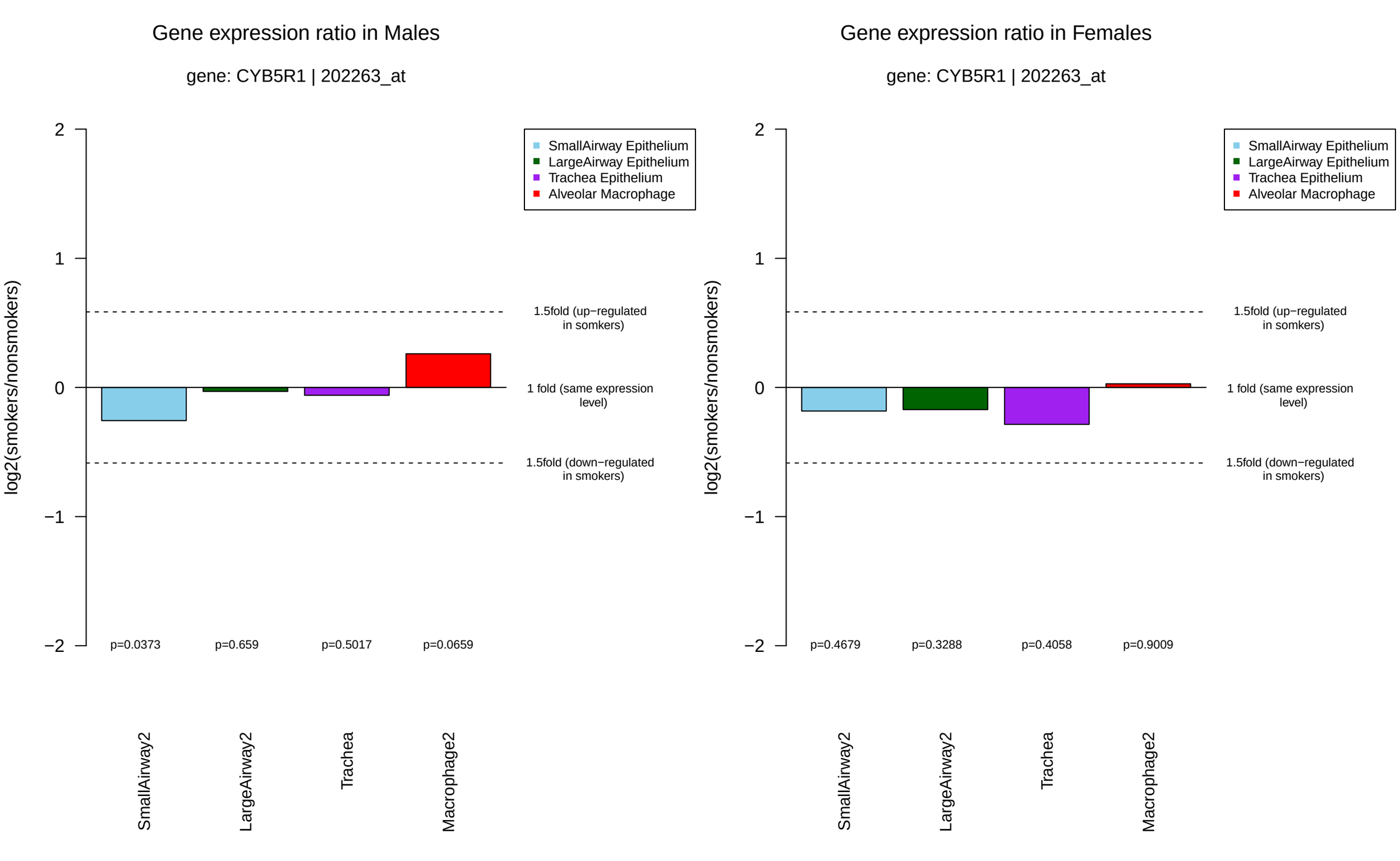
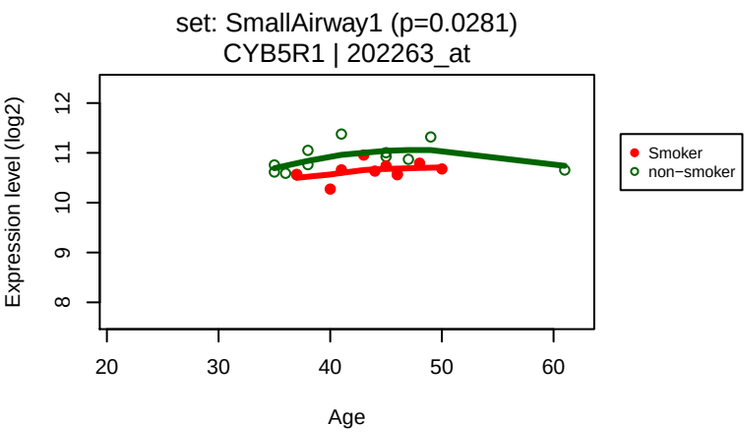
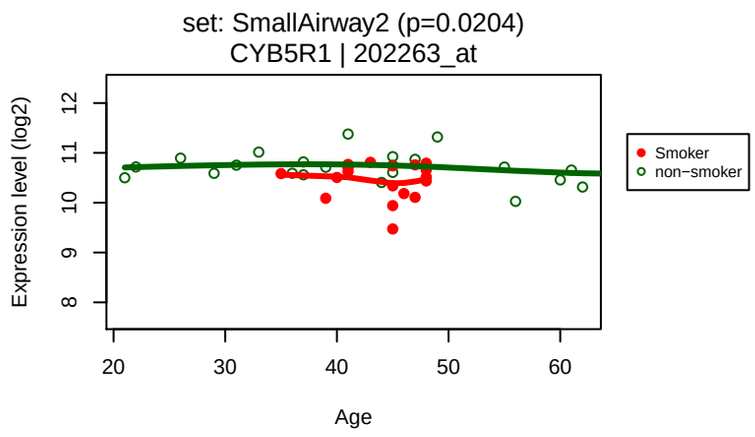
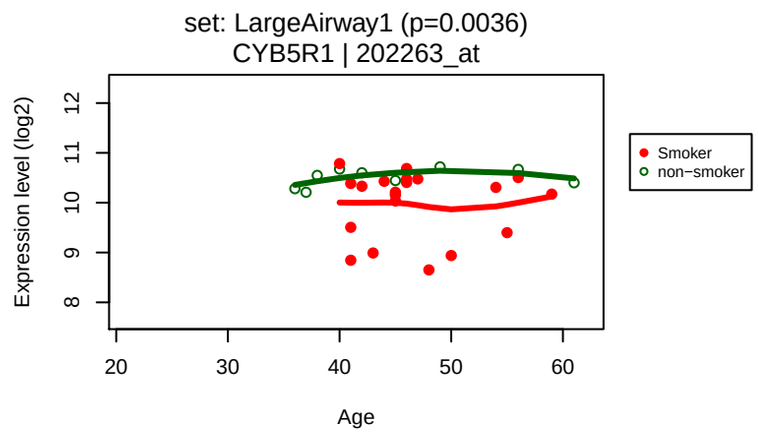

Gene expression of smokers and nonsmokers
gene: CYB5R1 | 202263_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0281
p=0.0204
p=0.0101
p=0.0036
p=0.314
p=0.2857
p=0.3421
p=0.1017
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

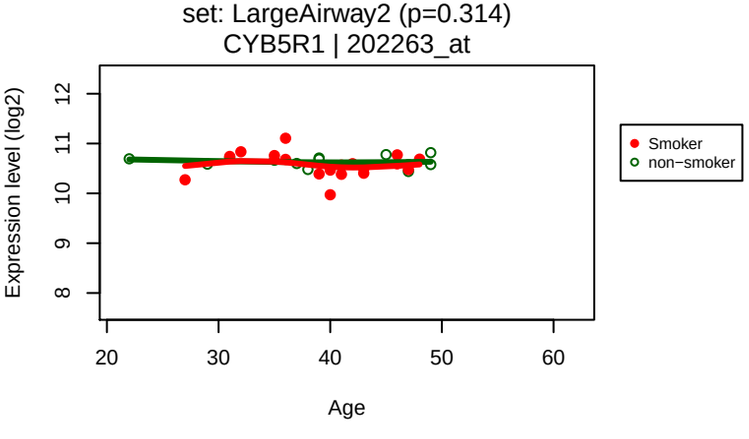

Gene expression ratio (smokers/nonsmokers)
gene: CYB5R1 | 202263_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0281
p=0.0204
p=0.0101
p=0.0036
p=0.314
p=0.2857
p=0.3421
p=0.1017
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

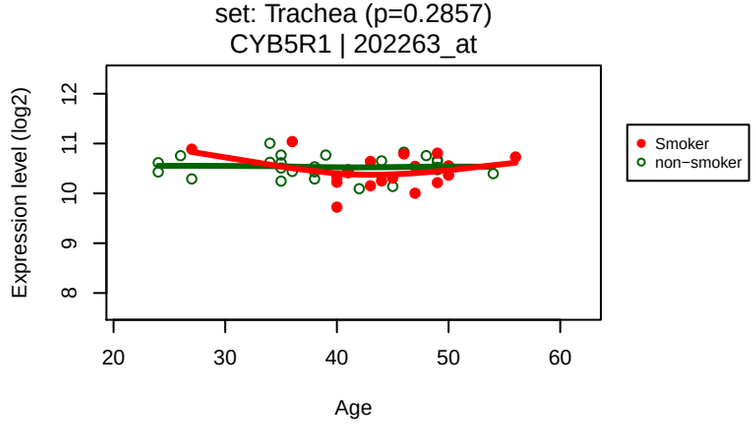

Gene expression ratio in Males
gene: CYB5R1 | 202263_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0373
p=0.659
p=0.5017
p=0.0659
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYB5R1 | 202263_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4679
p=0.3288
p=0.4058
p=0.9009
SmallAirway2
LargeAirway2
Trachea
Macrophage2

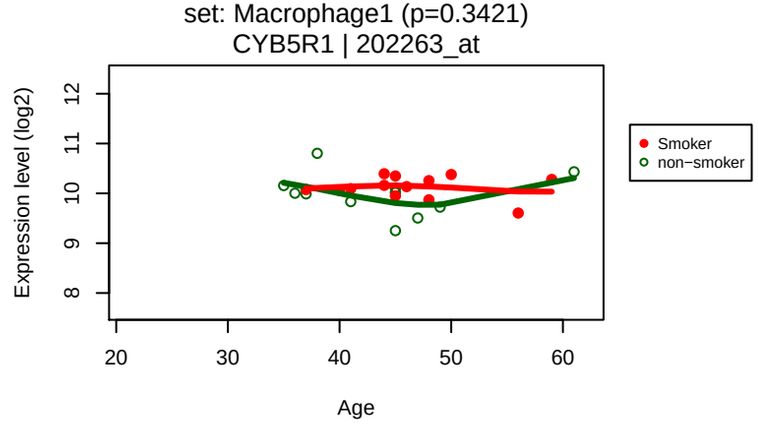

set: SmallAirway1 (p=0.0281)
CYB5R1 | 202263_at
Expression level (log2)
Age
Smoker
non−smoker

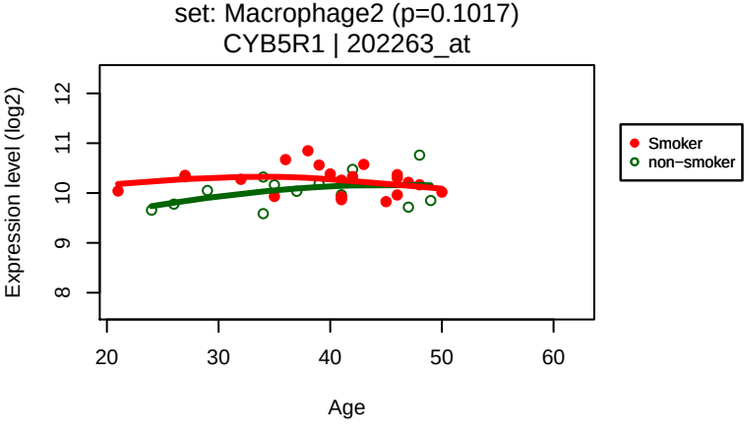

set: SmallAirway2 (p=0.0204)
CYB5R1 | 202263_at
Expression level (log2)
Age
Smoker
non−smoker

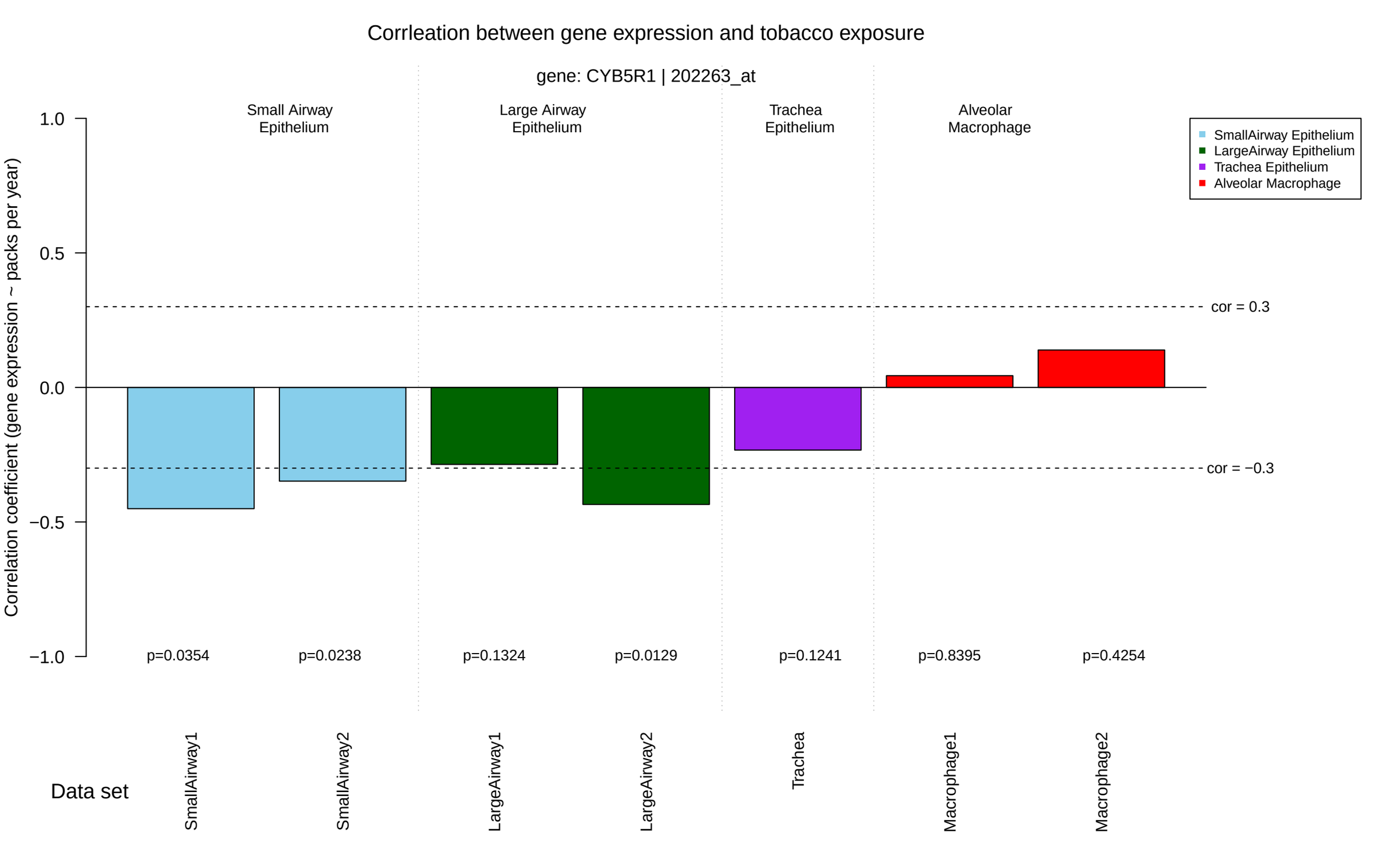

set: LargeAirway1 (p=0.0036)
CYB5R1 | 202263_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.314)
CYB5R1 | 202263_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2857)
CYB5R1 | 202263_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3421)
CYB5R1 | 202263_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1017)
CYB5R1 | 202263_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYB5R1 | 202263_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0354
p=0.0238
p=0.1324
p=0.0129
p=0.1241
p=0.8395
p=0.4254
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2